LANCERSNAP

Software Design Document (SDD)

CSCI 5520 - AI-Driven Software Development

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1 Problem & Users

1.1 Problem Description

Students at California Baptist University encounter lost items, facility problems, safety hazards, housing concerns, and IT issues across campus, but there is no single, simple process for reporting and following up on these different situations. This creates problems for both students and campus staff.

Key problems include:

- **Unclear reporting responsibility:** Students may not know which campus office should handle a particular issue, which can lead to reports being sent to the wrong place or not submitted at all.
- **Incomplete or inconsistent information:** Students may provide vague descriptions or miss important details, making it harder for staff to understand, prioritize, and act on a report.
- **Disconnected lost-and-found process:** Students who lose an item have limited ways to discover whether a matching item has been found, especially when the lost and found descriptions are different.
- **Privacy and safety concerns:** Photos submitted with reports can unintentionally expose sensitive information, while urgent safety hazards may be mixed with routine campus issues.
- **Duplicate and manual work for staff:** Multiple reports about the same problem can create unnecessary work, while staff may need to manually sort, route, and organize incoming reports.
- **Limited visibility after reporting:** Students may not know whether their report was received, assigned, or resolved, and the university has limited visibility into recurring issues across campus.

Core problem: CBU lacks a centralized reporting process that makes campus issues and lost-and-found cases easier to submit, manage, and follow through from initial report to resolution.

1.2 Target Users & Stakeholders

2 Proposed Solution & Scope

User / Stakeholder	Relationship to the Problem
CBU Student	Primary user who needs a quick and simple way to report campus issues or lost/found items without knowing which office is responsible.
Campus Office Staff	Receives and processes reports and needs clear, organized information to review, prioritize, and resolve them efficiently.
LancerSnap Administrator	Manages system access, reporting categories, office routing, and administrative configuration.
Safety Services	Handles safety-related concerns and needs potentially urgent hazards to be clearly identified and prioritized.
Facilities	Handles physical campus maintenance and infrastructure issues and needs reports to reach the correct queue with useful details.

Housing	Handles residential and housing-related concerns and needs relevant reports routed to the appropriate staff.
IT	Handles technology-related problems and needs reports categorized and directed to the appropriate support queue.
University Leadership	Uses aggregated reporting trends to understand recurring campus issues and identify opportunities for operational or infrastructure improvements.

2.1 Proposed Solution

LancerSnap is an AI-powered campus reporting and lost-and-found application designed to make reporting a campus issue as simple as taking a photo. Students can capture or upload one photo, add an optional note, and select or confirm the location without having to determine which campus office should handle the issue.

The system uses AI to analyze the submission, identify the report category, extract useful details, assess urgency, and route the report to the appropriate campus office queue. For lost-and-found reports, students can search using a description or photo, receive potential matches, and complete an ownership-verification step before making a claim.

LancerSnap also protects privacy by identifying and redacting sensitive information in images before they are shown in student-facing views. Students can track their reports from submission through resolution, while staff can manage incoming reports through office-specific dashboards.

The goal is to make campus reporting faster and easier for students, more organized for staff, and more useful for identifying recurring campus problems.

2.2 Minimum Viable Product (MVP) Scope

Student capabilities

- Sign in to LancerSnap using an authorized student account.
- Take or upload one photo for a report.
- Add an optional note.
- Select or confirm a CBU building or location.
- Submit a report without selecting the responsible campus office.
- Receive an AI-generated report category such as:
    - Lost item
    - Found item
    - Facilities issue
    - Safety hazard

    - IT problem

- View AI-extracted information such as the item/issue description, color, brand when available, location, and urgency.
- Receive a high-priority indication when a report is identified as a potential safety hazard.
- Have sensitive visual information automatically redacted before student-facing display.
- Track the status of submitted reports.
- Search for lost items using a natural-language description or photo.
- View potential matches for a lost item.
- Receive notifications when a potential match is identified.
- Complete an ownership-verification step before attempting to claim a matched item.

Staff capabilities:

- View reports routed to their authorized office queue.
- Review the report photo, student-provided information, and AI-generated details.
- Identify high-priority reports.
- Update report status.
- Review potential lost-and-found matches.
- View basic duplicate-report information when available.

Supporting capabilities:

- Secure authentication and role-based access.
- Storage for report information and uploaded images.
- Configurable report categories and office-routing rules.
- Notifications for important status changes and potential lost-item matches.
- Basic audit information for report creation and status changes.

2.3 Out of Scope / Non-Goals

The following are intentionally outside the current MVP:

- Direct integration with 911, police dispatch, fire/EMS, or other public emergency services.
- Replacing existing university systems used for full work-order or case management.
- AI making final decisions about safety incidents, ownership, discipline, or other administrative actions.
- Facial recognition or identifying individual people from uploaded photographs.
- Continuous GPS tracking of students.
- Automatically releasing or transferring physical lost-and-found property.

- Automatic disciplinary action based on submitted reports.
- Advanced predictive analytics or long-term campus forecasting.
- Full campus-wide integration with every university administrative system.
- Automated public posting of all submitted photographs.
- Native desktop applications.
- Automatic resolution of ambiguous or disputed reports without human review.

Core non-goal:

LancerSnap is not intended to replace human staff judgment or emergency services. AI assists with classification, information extraction, prioritization, and matching, while authorized personnel remain responsible for final operational decisions.

3 User Stories & Acceptance Criteria

The following user stories define the major behaviours required for the LancerSnap MVP. Each story uses a stable identifier and includes an observable pass/fail condition so that the behaviour can be evaluated during implementation and testing.

3.1 Student

US-01 – Submit Report

User Story: As a student, I can submit a photo with an optional note and location so that I can quickly report a campus issue or lost/found item.

Pass/Fail Check**:** Pass if the report is submitted and a confirmation is displayed.

US-02 – AI Classify Report

User Story: As a student, I can see the category assigned to my report so that I know how it was interpreted.

Pass/Fail Check: Pass if the report receives a supported category.

US-03 – Protect Photo Privacy

User Story: As a student, I can submit a photo without unnecessarily exposing sensitive information.

Pass/Fail Check: Pass if configured sensitive information is redacted before student-facing display.

US-04 – Search Lost Items

User Story: As a student, I can search for a lost item using a description or photo so that I can find possible matches.

Pass/Fail Check: Pass if the system returns zero or more potential matches.

US-05 – Verify Ownership

User Story: As a student, I can verify ownership of a matched item so that unauthorized claims are reduced.

Pass/Fail Check: Pass if the claim requires the configured verification step.

US-06 – Track Report

User Story: As a student, I can view my report status so that I know its progress.

Pass/Fail Check: Pass if the current status is displayed and updated after an authorized change.

3.2 Campus Staff

US-07 – View Assigned Reports

User Story: As campus staff, I can view reports assigned to my office so that I can process them.

Pass/Fail Check: Pass if staff can access their office's reports only.

US-08 – Review AI Information

User Story: As campus staff, I can review AI-generated classification and details so that I can make

informed decisions.
Pass/Fail Check: Pass if the AI-generated information is visible in the report.

US-09 – Update Report Status

User Story: As campus staff, I can update a report's status so that the student can track its progress.
Pass/Fail Check: Pass if the updated status is saved and visible to the student.

3.3 Administrator

US-10 – Manage Routing

User Story: As an administrator, I can configure report categories and office routing so that reports reach the correct team.
Pass/Fail Check: Pass if a saved routing rule correctly determines the destination office.

US-11 – View Report Trends

User Story: As an administrator, I can view aggregated report trends so that recurring campus issues can be identified.
Pass/Fail Check: Pass if reports can be grouped by at least one supported category, location, or time period.

4 Requirements, Constraints & Design Rules

RC-01 – Privacy: Uploaded images must not expose sensitive information to unauthorized users. Sensitive content identified for redaction must be protected before student-facing display.

RC-02 – Access Control: Students may access only their own reports, while staff may access reports assigned to their authorized office. Administrative functions require administrator access.

RC-03 – AI Uncertainty: AI-generated classifications, extracted details, urgency levels, and matches must be presented as recommendations when uncertainty exists and must not be treated as guaranteed facts.

RC-04 – Human Oversight: Final decisions involving safety, ownership, report resolution, or administrative action must remain with authorized human staff.

RC-05 – Safety Priority: Reports identified as potential safety hazards must be clearly marked as high priority so they are distinguishable from routine reports.

RC-06 – Secure Media: Uploaded photos must be stored using access controls and must not be publicly accessible by default.

RC-07 – Error Handling: If AI processing, image analysis, or notifications fail, the system must clearly indicate the failure and must not report a false successful result.

RC-08 – Accessibility: Core reporting and tracking functions must use clear labels, readable text, sufficient contrast, and must not rely on color alone to communicate important information.

RC-09 – Configurable Routing: Report-to-office routing must be configurable by an authorized administrator rather than permanently fixed in the interface.

RC-10 – Emergency Boundary: LancerSnap must clearly state that it is a campus reporting tool and does not replace contacting emergency services when immediate emergency assistance is required.

5 Platform & Architecture

5.1 System Components & Technology Choices

LancerSnap will use a mobile-first architecture consisting of a mobile application, cloud backend, database and storage services, an AI processing layer, and notification/location services.

Technology / Platform	Rationale	Usage	Residency
React Native + Expo	Supports rapid development of a cross-platform mobile application using a single codebase.	Student, staff, and administrator mobile interfaces.	User mobile device.
FastAPI (Python)	Provides a lightweight backend suitable for API development and AI integration.	Handles reports, routing, permissions, status updates, and AI orchestration.	Cloud-hosted backend.
PostgreSQL	Provides reliable structured storage for users, reports, statuses, routing rules, and matches.	Stores core application data.	Cloud database.
Supabase	Provides managed PostgreSQL, authentication, storage, and backend services in one platform, reducing infrastructure complexity for the prototype.	Authentication, database access, and protected file storage.	Cloud-hosted service.
AI Multimodal Model	Can analyze images and text to classify reports and extract useful information.	Report classification, attribute extraction, urgency assessment, and lost-and-found analysis.	Cloud AI service.
Vector Similarity Search	Allows lost-item descriptions and images to be compared based on similarity rather than exact keywords.	Finds potential lost-and-found matches.	Cloud database/backend.
Object Storage	Separates uploaded images from application records and supports access control.	Stores report and lost-and-found photographs.	Cloud storage.
Push Notification Service	Provides timely notifications without requiring students to repeatedly check the application.	Lost-item matches and report-status updates.	Cloud notification service and user device.

Device Location Services	Allows users to identify where an issue or item was encountered.	Building/location selection and optional device location.	User mobile device.

Major System Components:

Mobile Application: Provides the primary student reporting, lost-and-found, status tracking, and staff interfaces.

Backend API: Handles report creation, authentication context, routing, business rules, and communication between the application and other services.

AI Processing Layer: Analyzes submitted text and images, generates classifications and extracted attributes, identifies potential urgency, and supports lost-and-found matching.

Database & Storage: Stores structured report information, user data, statuses, routing rules, match information, and uploaded photographs.

Notification & Location Services: Provide report updates, match notifications, and location information.

5.2 Architecture Diagram & Explanation

- **Mobile Application:** Students, staff, and administrators use the LancerSnap mobile application to submit and manage reports.
- **Backend API:** The backend receives requests from the mobile app and manages report submission, routing, permissions, and status updates.
- **AI Processing:** Submitted photos and notes are analysed by the AI to classify the report, extract useful details, assess urgency, and support lost-and-found matching.
- **Database:** User accounts, reports, categories, statuses, routing rules, and match information are stored in the application database.
- **Image Storage:** Uploaded photographs are stored separately with access controls to protect potentially sensitive information.
- **Smart Routing:** After AI classification, the backend uses routing rules to send the report to the appropriate campus office, such as Facilities, Safety Services, Housing, or IT.
- **Lost-and-Found Matching:** The matching component compares lost-item descriptions or images with found-item reports and returns potential matches.
- **Notifications:** Students receive notifications for important events such as potential lost-item matches and report-status updates.
- **Location Services:** Students can select or provide their campus location when submitting a report.
- **Human Oversight:** AI provides recommendations and extracted information, while authorized staff remain responsible for final decisions about safety, ownership, and resolution.

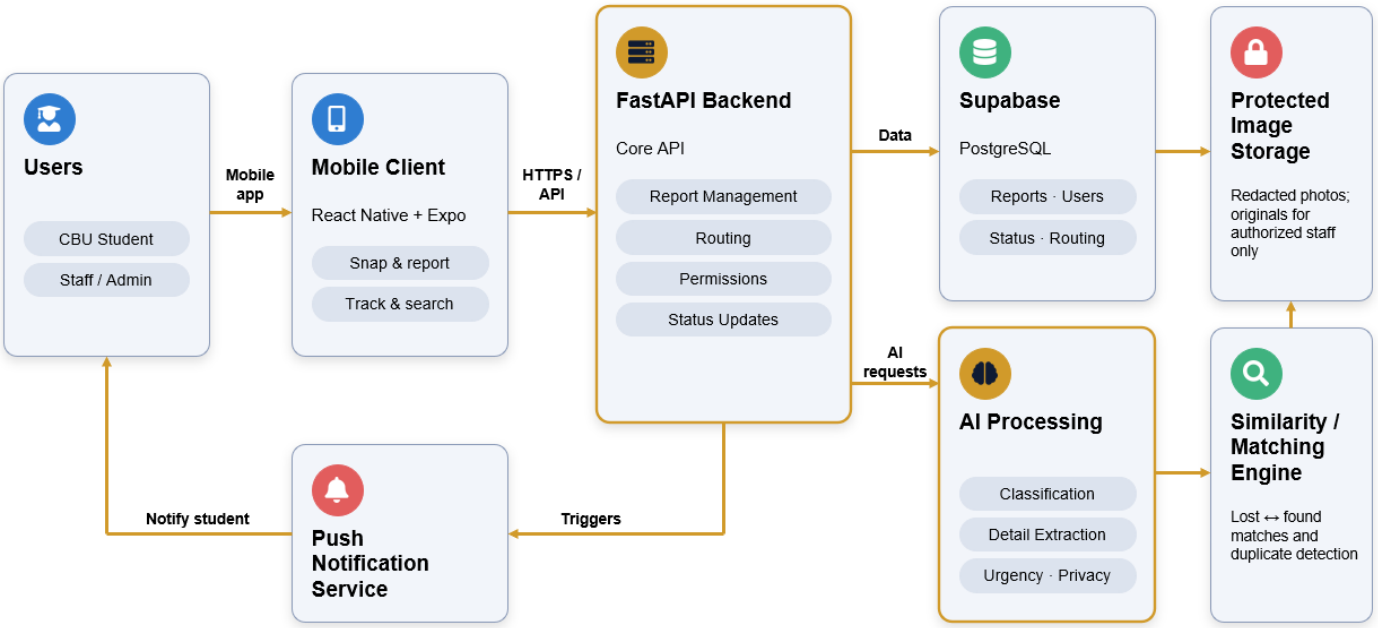


Figure 1: LancerSnap System Architecture and Data Flow

6 UI/UX Design

6.1 Application / Experience Flow

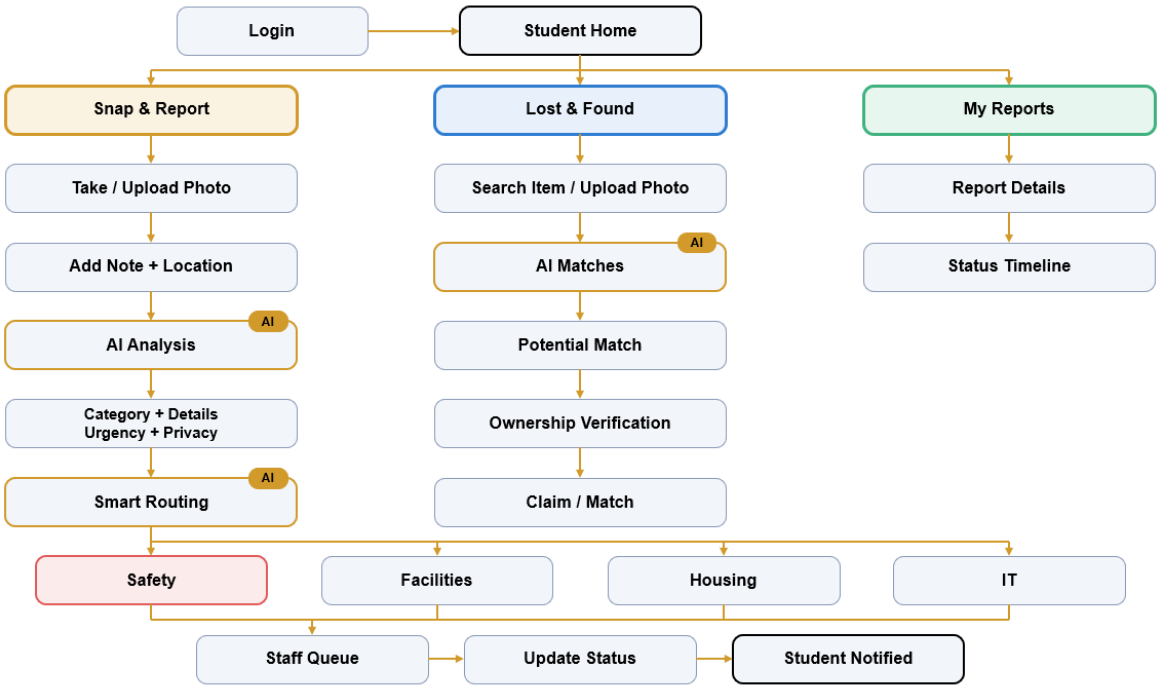


Figure 2:LancerSnap Application / Experience Flow.

LancerSnap uses a role-based mobile experience centered around two primary student tasks: reporting campus issues and searching for lost items. After signing in, students reach the home screen where they can quickly access the reporting, lost-and-found, and report-tracking features.

The main reporting flow is **Home → Snap & Report → Photo/Note/Location → AI Analysis → Smart Routing → Staff Queue → Status Updates**. The lost-and-found flow is **Home → Lost & Found → Description/Photo → Potential Matches → Ownership Verification**. Students can also access **My Reports** to view the current status of previously submitted reports.

Authorized staff follow a separate flow from **Staff Dashboard → Report Queue → Report Details → Review AI Information → Update Status**.

6.2 Interactive UI/UX Prototype

LancerSnap was developed as a mobile-first interactive prototype using Figma Make. The prototype represents the major student and staff experiences defined in the MVP, including campus reporting, AI-assisted analysis, lost-and-found matching, ownership verification, report tracking, and staff report management.

The prototype supports the following primary interactions:

- **Snap & Report:** Students capture or upload a photo, add an optional note and location, review AI analysis, and submit the report.
- **Lost & Found:** Students describe or upload a lost item, view potential matches, and complete ownership verification.
- **Report Tracking:** Students open a submitted report and view its current status and staff updates.
- **Staff Processing:** Staff review routed reports, view AI-generated information, and update report status.

Prototype / Figma Make Link: https://www.figma.com/proto/makzxHRmehtu2jH9pJ4GZ6/Untitled?node-id=3-19147&t=n9EpO3Erz7dkIOce-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1

Required interactive flows to be finalized:

1. Snap & Report
2. Lost & Found Matching
3. Report Status Tracking

6.3 Design Alignment & Human Refinement

The initial LancerSnap prototype was reviewed against the user stories and requirements defined in Sections 3 and 4. The current design reflects the photo-first reporting workflow, AI-assisted classification, privacy protection, lost-and-found matching, ownership verification, and report tracking requirements.

Further human-directed refinement is pending the next available Figma Make generation cycle. The planned refinement review will focus on:

Design Refinement	Why It Will Be Reviewed	Requirement
Emphasize the Snap & Report action	Ensure the primary reporting action is immediately visible to students.	US-01
Clarify AI-generated information and confidence	Prevent AI suggestions from appearing to be confirmed facts.	US-02 / RC-03
Strengthen privacy feedback during image analysis	Make the protection of sensitive information visible to students.	US-03 / RC-01
Clarify potential lost-item matches	Ensure similarity results are understood as recommendations rather than proof of ownership.	US-04 / US-05 / RC-04

Note: These refinements will be implemented and documented after the Figma Make limit resets.

7 Key Design Decisions & Open Questions

7.1 Key Design Decisions

DD-01 – Photo-First Reporting: LancerSnap uses a photo-first reporting process to reduce the effort required for students to submit campus issues.

DD-02 – AI-Assisted Decisions: AI will classify reports, extract details, identify urgency, and recommend lost-and-found matches, but authorized staff will make final operational decisions.

DD-03 – Smart Routing: Reports will be routed using configurable category-to-office rules so students do not need to know which office is responsible.

DD-04 – Privacy Before Sharing: Sensitive information in uploaded images will be protected before student-facing display.

DD-05 – Ownership Verification: A potential lost-and-found match will not automatically confirm ownership; an additional verification step is required before a claim can proceed.

DD-06 – Human Review for Uncertainty: When AI confidence is low or information is ambiguous, the system will indicate uncertainty rather than presenting the result as a confirmed fact.

7.2 Open Questions, Assumptions & Risks

OQ-01 – AI Provider: The final AI model/provider has not yet been selected and will be evaluated based on accuracy, privacy, cost, and response time.

OQ-02 – Image Matching: The final method for visual lost-and-found matching still needs to be evaluated using representative item images.

OQ-03 – Notification Method: The final notification method for status updates and potential matches has not yet been selected.

OQ-04 – Lost-and-Found Workflow: The appropriate CBU office or operational process for handling found property needs to be confirmed.

RISK-01 – Incorrect AI Classification: AI may assign an incorrect category or urgency level. Staff review and uncertainty indicators will reduce this risk.

RISK-02 – False Matches: Visually similar items may not belong to the same person. Ownership verification is required before a claim proceeds.

RISK-03 – Privacy Redaction Failure: Automated redaction may not detect every sensitive element in an image. Original images must therefore remain access-controlled.

RISK-04 – AI Service Availability: Dependence on external AI services may cause delays or failures if the service is unavailable or rate-limited.

RISK-05 – Safety Misclassification: A serious hazard could be classified incorrectly. Safety-related reports should therefore receive conservative priority handling and remain subject to human review.

8 References & Supporting Links (Optional)

- California Baptist University - Official university website and campus information.
  https://calbaptist.edu/
- Figma - Figma Make documentation.
  https://help.figma.com/
- React Native - React Native documentation.
  https://reactnative.dev/
- FastAPI - FastAPI documentation.
  https://fastapi.tiangolo.com/
- Supabase - Supabase documentation.
  https://supabase.com/docs